

$$\begin{aligned} \text{111. } 5358 \times 51 &= 5358 \times (50 + 1) = 5358 \times 50 + 5358 \times 1 \\ &= \frac{5358 \times 100}{2} + 5358 \\ &= \frac{535800}{2} + 5358 = 267900 + 5358 = 273258. \end{aligned}$$

112. $587 \times 999 = 587 \times (1000 - 1) = (587 \times 1000) - (587 \times 1)$
 $= 587000 - 587 = 586413.$

113. $3897 \times 999 = 3897 \times (1000 - 1) = (3897 \times 1000) - (3897 \times 1)$
 $= 3897000 - 3897 = 3893103.$

114. $72519 \times 9999 = 72519 \times (10000 - 1)$
 $= (72519 \times 10000) - (72519 \times 1)$
 $= 725190000 - 72519 = 725117481.$

$$\begin{array}{r} 725190000 \\ - 72519 \\ \hline 725117481 \end{array}$$

115. $2056 \times 987 = 2056 \times (1000 - 13)$
 $= (2056 \times 1000) - (2056 \times 13)$
 $= 2056000 - 26728 = 2029272.$

116. $1904 \times 1904 = (1904)^2 = (1900 + 4)^2$
 $= (1900)^2 + 4^2 + 2 \times 1900 \times 4$
 $= 3610000 + 16 + 15200 = 3625216.$

117. $1397 \times 1397 = (1397)^2 = (1400 - 3)^2$
 $= (1400)^2 + 3^2 - 2 \times 1400 \times 3 = 1960000 + 9 - 8400$
 $= 1951609.$

118. $(107 \times 107) + (93 \times 93) = (107)^2 + (93)^2$
 $= (100 + 7)^2 + (100 - 7)^2$
 $= (a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
 $= 2[(100)^2 + 7^2] = 2[10000 + 49]$
 $= 2 \times 10049 = 20098.$

119. $(217 \times 217) + (183 \times 183) = (217)^2 + (183)^2$
 $= (200 + 17)^2 + (200 - 17)^2 = (a + b)^2 + (a - b)^2$
 $= 2(a^2 + b^2)$, where $a = 200$, $b = 17$
 $= 2 [(200)^2 + (17)^2] = 2 [40000 + 289]$
 $= (2 \times 40289) = 80578.$

120. $(106 \times 106 - 94 \times 94) = (106)^2 - (94)^2$
 $= (106 + 94)(106 - 94) = (200 \times 12) = 2400.$

$$\begin{aligned} \text{121. } (8796 \times 223 + 8796 \times 77) &= 8796 \times (223 + 77) \\ &\quad \text{[by distributive law]} \\ &= (8796 \times 300) = 2638800. \end{aligned}$$

$$\begin{aligned} 122. \quad & (287 \times 287 + 269 \times 269 - 2 \times 287 \times 269) \\ &= (287)^2 + (269)^2 - (2 \times 287 \times 269) \\ &= (287 - 269)^2 = (18)^2 = 324. \end{aligned}$$

123. $(476 + 424)^2 - 4 \times 476 \times 424 = (a + b)^2 - 4ab = (a - b)^2$
 where $a = 476$ & $b = 424 = (476 - 424)^2 = (52)^2 = (50 + 2)^2$
 $= (50)^2 + 2^2 + 2 \times 50 \times 2$
 $= (2500 + 4 + 200) = 2704.$

124. $(112 \times 5^4) = 112 \times \left(\frac{10}{2}\right)^4 = \frac{112 \times 10000}{16} = 70000.$

$$\begin{aligned} \text{125. } (5746320819 \times 125) &= \frac{5746320819 \times (125 \times 8)}{8} \\ &= \frac{5746320819 \times 1000}{8} = \frac{5746320819000}{8} = 718290102375. \end{aligned}$$

$$\begin{aligned} \text{126. } 935421 \times 625 &= \frac{935421 \times 625 \times 16}{16} = \frac{935421 \times 10000}{16} \\ &= \frac{9354210000}{16} = 584638125. \end{aligned}$$

127. $(999)^2 - (998)^2 = (999 + 998)(999 - 998) = (1997 \times 1) = 1997.$

128. $(80)^2 - (65)^2 + 81 = (80 + 65)(80 - 65) + 81 = (145 \times 15) + 81 = (2175 + 81) = 2256.$

129. $(24 + 25 + 26)^2 - (10 + 20 + 25)^2 = (75)^2 - (55)^2$
 $= (75 + 55)(75 - 55) = (130 \times 20) = 2600.$

130. $(65)^2 - (55)^2 = (65 + 55)(65 - 55) = (120 \times 10) = 1200.$

131. $(a^2 - b^2) = 19 \Rightarrow (a + b)(a - b) = 19$.
Clearly, $a = 10$ and $b = 9$.

132. Since $(a + b)^2 - (a - b)^2 = 4ab$, so the given expression should be a multiple of 4.

So, the least value of $4ab$ is 32 and so the least value of ab is 8.

Hence, the smallest value of a is 4 and that of b is 2.
Hence, $a = 4$.

133. Given Expression = $(397)^2 + (104)^2 + 2 \times 397 \times 104$
 $= (397 + 104)^2 = (501)^2 = (500 + 1)^2$
 $= (500)^2 + 1^2 + 2 \times 500 \times 1 = 250000 + 1 + 1000 = 251001.$

134. $(64)^2 - (36)^2 = 20 \times x \Rightarrow (64 + 36)(64 - 36) = 20 \times x$
 $\Rightarrow x = \frac{100 \times 28}{20} = 140.$

135. Given Expression

$$= \frac{(a+b)^2 - (a-b)^2}{ab} \text{ (where } a = 489, b = 375) = \frac{4ab}{ab} = 4$$

136. Given Expression = $\frac{(a+b)^2 + (a-b)^2}{(a^2 + b^2)}$,
(where $a = 963$ and $b = 476$) = $\frac{2(a^2 + b^2)}{(a^2 + b^2)} = 2$.

137. Given Expression

$$= \frac{(a^3 + b^3)}{(a^2 - ab + b^2)}, (\text{where } a = 768 \text{ and } b = 232) = (a + b)$$
$$= (768 + 232) = 1000.$$

$$\begin{aligned} &= (768 + 232) = 1000. \\ \text{138. Given Expression} &= \frac{(854)^3 - (276)^3}{(854)^2 + (854 \times 276) + (276)^2} \\ &= \frac{(a^3 - b^3)}{(a^2 + ab + b^2)}, \text{ where } a = 854 \text{ and } b = 276 \\ &= (a - b) = (854 - 276) = 578. \end{aligned}$$

139. Given Expression = $\frac{(753)^2 + (247)^2 - (753 \times 247)}{(753)^3 + (247)^3}$

$$= \frac{(a^2 + b^2 - ab)}{(a^3 + b^3)}, \text{ where } a = 753 \text{ and } b = 247$$
$$= \frac{1}{(a + b)} = \frac{1}{(753 + 247)} = \frac{1}{1000}.$$

140. Given Expression = $\frac{(256)^2 - (144)^2}{112} = \frac{(256 + 144)(256 - 144)}{112}$
 $= \frac{(400 \times 112)}{112} = 400.$

$$141. \frac{(a^2 + b^2 + ab)}{(a^3 - b^3)} = \frac{a^2 + b^2 + ab}{(a - b)(a^2 + b^2 + ab)} = \frac{1}{(a - b)} = \frac{1}{(11 - 9)} = \frac{1}{2}.$$

142. $a + b + c = 0 \Rightarrow a + b = -c, (b + c) = -a$ and $(c + a) = -b$
 $\Rightarrow (a + b)(b + c)(c + a) = (-c) \times$
 $(-a) \times (-b) = -(abc).$

$$\begin{aligned} \text{143. } (a^2 + b^2 + c^2 - ab - bc - ca) &= (a + b + c)^2 - 3(ab + bc + ca) \\ &= (7 + 5 + 3)^2 - 3(35 + 15 + 21) \\ &= (15)^2 - 3 \times 71 \\ &= (225 - 213) = 12. \end{aligned}$$

144. Both addition and multiplication of numbers are commutative and associative.